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GATE ECE · ANALOG CIRCUITS · CHAPTER-WISE PYQS

Oscillators

GATE ECE Analog Circuits · Previous Year Questions with Answer Key

8	8	1994 – 2016	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

INSIDE THIS PDF

Marks 1	1 question
Marks 2	6 questions
Marks 5	1 question
Answer key	at the end

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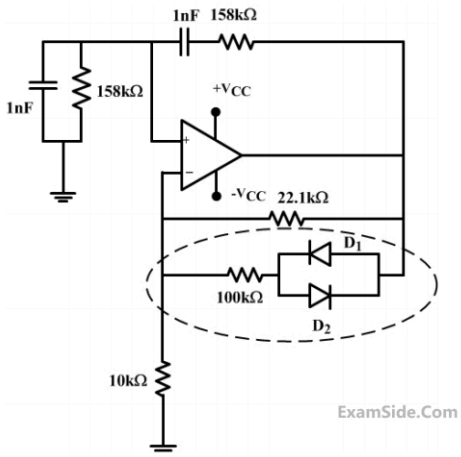
1. **Solve**
   Print it, or work through it on screen.
2. **Check**
   The answer key is on the last pages.
3. **Understand**
   Scan the QR code beside any question for its full solution.

Marks 1 1 questions

Q1. GATE ECE 2016 Set 1

Open solution

Consider the oscillator circuit shown in the figure. The function of the network (shown in dotted lines) consisting of the 100 k Ω resistor in series with the two diodes connected back- to is to:



(A) introduce amplitude stabilization by preventing the op-amp from saturating and thus producing sinusoidal oscillations of fixed amplitude

(B) introduce amplitude stabilization by forcing the op-amp to swing between positive and negative saturation and thus producing square wave oscillations of fixed amplitude

(C) introduce frequency stabilization frequency stabilization by forcing the circuit to oscillate at a single frequency

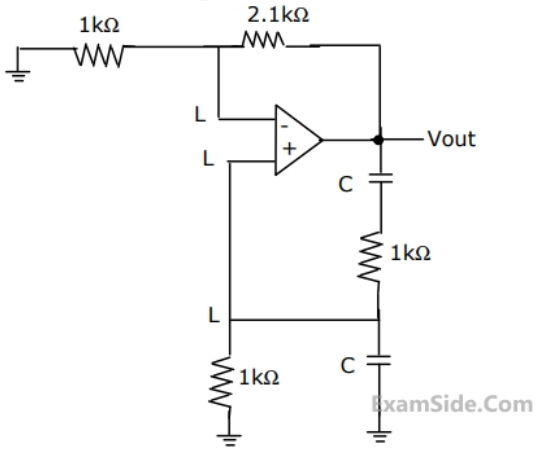
(D) enable the loop gain to take on a value that produces square wave oscillations

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Marks 2 6 questions

Q2. GATE ECE 2004

The value of 'C' required for a sinusoidal oscillation of frequency 1 KHz in the circuit of the figure is



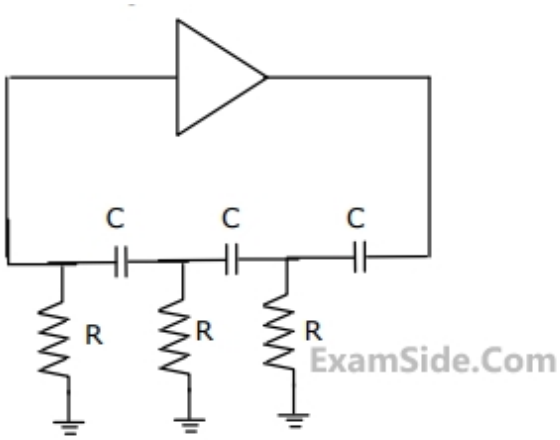
Open solution

(A) $\frac{1}{2\pi}\mu F$ (B) $2\pi\,\mu\,F$

(C) $\frac{1}{2\pi\sqrt{6}}\mu F$ (D) $2\pi\sqrt{6}\,\mu F$

Q3. GATE ECE 2003

The oscillator circuit shown in the figure has an ideal inverting Amplifier. It's frequency of Oscillation (in Hz) is



Open solution

(A) $\frac{1}{2\pi\sqrt{6}\,RC}$ (B) $\frac{1}{2\pi RC}$ (C) $\frac{1}{\sqrt{6}RC}$ (D) $\frac{1}{\sqrt{6}(2\pi RC)}$

Q4. GATE ECE 2002

The circuit in the figure employs positive Feedback and is intended to generate sinusoidal oscillation. If at a frequency f_0, B(f)= $\frac{\Delta V_f(f)}{V_0(f)} = \frac{1}{6}\angle 0^0$ then to sustain oscillation at this frequency.

Open solution

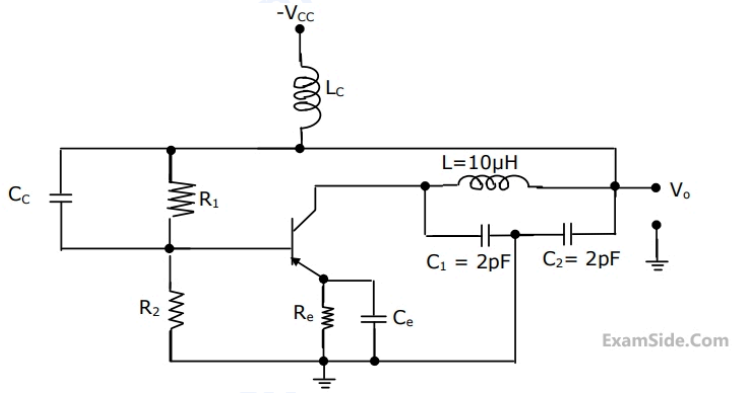
(A) $R_2 = 5\ R_1$ (B) $R_2 = 6\ R_1$ (C) $R_2 = \frac{R_1}{5}$ (D) $R_2 = \frac{R_1}{6}$

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Q5. GATE ECE 2001

Open solution

The Oscillator circuit shown in the figure is



(A) Hartley Oscillator with $f_{oscillation}$ = 79.6 MHz

(B) Colpitts Oscillator with $f_{oscillation}$ =50.3 MHz

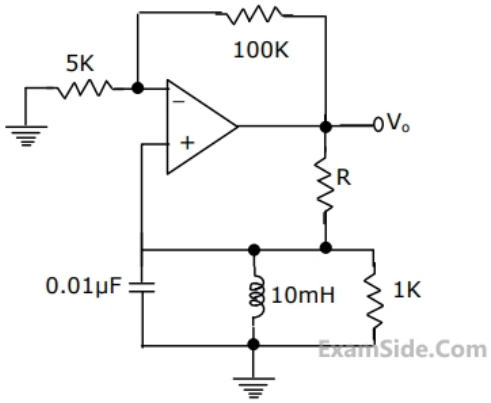
(C) Hartley Oscillator with $f_{oscillation}$ = 159.2 MHz

(D) Colpitts Oscillator with $f_{oscillation}$=159.2 MHz

Q6. GATE ECE 1996

Open solution

Value of R in the oscillator shown in the given figure. So chosen that it just oscillates at an angular frequencies of' "ω". The value of "ω" and the required value of R will respectively be



(A) 10^5 rad/ sec, $2 \times 10^4\ \Omega$

(B) 2×10^4 rad/ sec, $2 \times 10^4\ \Omega$

(C) 2×10^4 rad/ sec , $10^5\ \Omega$

(D) $10^5\ rad/\sec, 10^5\ \Omega$

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Q7. GATE ECE 1994

Open solution

Match the following.

GROUP-1

(A) Hartley

(B) Wein-bridge

(C) Crystal

GROUP-2

(1) Low frequency oscillator

(2) High frequency oscillator

(3) Stable frequency oscillator

(4) Relaxation oscillator

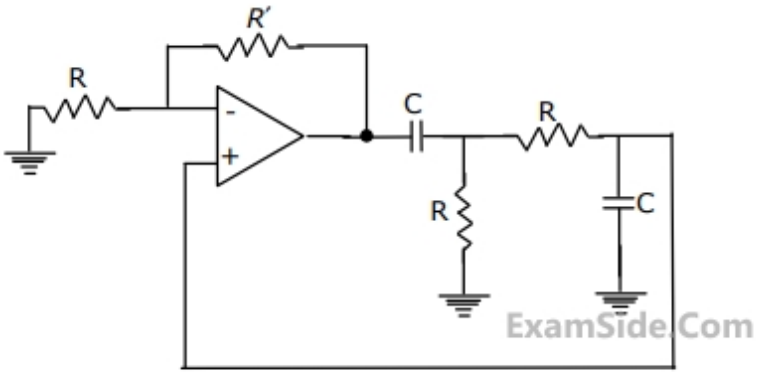
(5) Nagative Resistance oscillator

Marks 5 1 questions

Q8. GATE ECE 1998

Open solution

Find the value of R^1 in the circuit of fig. For generating sinusoidal Oscillations. Find the frequency of oscillations.



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Answer Key

Marks 1

1. **A**				

Marks 2

2. **A**	3. **A**	4. **A**	5. **B**	6. **A**
7. **(A)-(2), (B)-(1), (C)-(3)**				

Marks 5

8. R^1 **= 2R** **Frequency of oscillation W =** $\frac{1}{CR}$

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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